

WORK EXPERIENCE

AI/ML Computational Science Specialist, Accenture

09/2026 – Present

- **Forensic Knowledge Assistant (Multimodal RAG):** Designed a citation-grounded Q&A assistant (Microsoft Foundry, Azure AI Search, Azure Functions, Copilot Studio, Power Platform) answering engineering-investigation questions from text, an image, or both; enforced grounding in code under a cite-or-silent policy, secured with Entra ID; recall@10 0.84, RAGAS factual correctness 0.87, 2% false-answer / 5% false-refusal, p95 8s text-only / 20s with image.

AI/ML Computational Science Sr Analyst, Accenture

09/2024 – 08/2026

- **Privacy-Safe Analytics Agent:** Architected an Azure analytics platform — incremental ingestion with PII/profanity redaction, hybrid BM25 + vector AI Search over 4M+ source records consolidated into 100K+ indexed documents, four-agent LangGraph reasoning layer — shipped via Docker/GitHub Actions CI/CD with OpenTelemetry and Application Insights tracing; routing accuracy 97%, query-parsing 81%, retrieval precision 0.91 / recall 0.92; cutting time-to-insight from hours to seconds (~95% faster), retrieval latency ~75%, analyst effort 60–70%.
- **Enterprise Data Q&A Agent:** Engineered a multi-agent text-to-SQL system (Copilot Studio, Microsoft Foundry, Azure Functions) translating natural-language questions into T-SQL against Fabric Lakehouse endpoints, with SELECT-only validation and an error-feedback self-correction loop; 10M+ records in <30s, routing accuracy 94%.
- **GenAI Business Doc Generator:** Delivered a multi-agent RAG system with LangGraph + LoRA fine-tuned LLM for business document generation; LangSmith tracing, ROUGE evals; cutting drafting effort 70–80% at 100+ documents/week.

Software Developer, Expro

09/2022 – 09/2024

- **Well Performance Analytics Platform:** Implemented a PyTorch LSTM-autoencoder pipeline for anomaly detection on well-sensor time-series data, improving risk detection by ~20% and reducing analysis time by ~40%.
- **Oil & Gas Insights Web Application:** Designed a PySpark KPI pipeline on Azure Databricks using PCA, K-means, and Random Forest to derive KPI features from equipment telemetry, reducing manual reporting by ~50%.

Data Scientist, Ganit Labs

03/2021 – 09/2022

- **Equipment Failure Prediction Model:** Developed predictive maintenance models on GCP for oilfield equipment with ~85% recall and ~30% less downtime.

Biomedical Engineer, Portea, AOV, Care Group

08/2018 – 09/2020

- Led cross-functional teams in medical equipment maintenance and training programs.

EDUCATION

MTech (Master of Technology), University of Calcutta

2016 – 2018

BTech (Bachelor of Technology), WBUT

2011 – 2015

TECHNICAL SKILLS

Languages:

Python, SQL

AI Architecture & Governance:

Solution architecture, multi-agent & RAG design patterns, AI guardrails, citation grounding, human-in-the-loop governance, PII redaction, cost/latency trade-off analysis

Cloud Platforms:

Azure (AI Foundry, Storage, Fabric, Cosmos DB, AI Search, OpenAI, App Service, Function App, Logic Apps, Container Apps); GCP (Cloud Storage, BigQuery, App Engine)

MLOps & DevOps:

Docker, Git, GitHub Actions CI/CD, Azure DevOps, Bicep IaC, MLOps & LLMops pipelines, LangSmith, OpenTelemetry, Application Insights (KQL), OpenEvals, RAGAS, REST APIs

Generative AI & Frameworks:

Multi-agent orchestration, Retrieval-Augmented Generation (multimodal & hybrid BM25 + vector), Prompt engineering, LLM fine-tuning (LoRA), Observability, Evaluation, LangChain, LangGraph, CrewAI, Claude Agent SDK, MCP, Microsoft Foundry Agents, Copilot Studio, Power Platform, Power Automate

Machine Learning & Deep Learning:

LSTM autoencoders, Isolation Forest, PCA, K-means, HDBSCAN, Random Forest, Decision Tree, Linear Regression, Logistic Regression, NLP

Data & Retrieval:

Vector search: FAISS, Pinecone, Azure AI Search · Graph: Neo4j · Platforms: Microsoft Fabric, Azure SQL, Azure Databricks

Frameworks & Libraries:

PyTorch, scikit-learn, Hugging Face Transformers, Azure OpenAI / OpenAI SDK, FastAPI, Flask

CERTIFICATIONS & TRAININGS

• [Claude Certified Architect – Foundations](#)

• [AZ-900: Microsoft Azure Fundamentals](#)

• [AI-102: Microsoft Azure AI Engineer Associate](#)

• [AI-900: Microsoft Azure AI Fundamentals](#)

• [CS50AI: Introduction to Artificial Intelligence with Python — Harvard University](#)

• [LLMOps, MLOps & CI/CD/CT Pipelines \(Udemy\)](#)

PROJECTS & PUBLICATIONS

[rti-engine – AI Pay Transparency Compliance Engine](https://github.com/trayan4/rti-engine), <https://github.com/trayan4/rti-engine>

Architected an 11-node LangGraph multi-agent system for pay-transparency compliance — 6 model-driven, 5 deliberately deterministic where no judgment is required — with tiered autonomy, human approval gates, cross-vendor review, figure-validation. MCP-exposed Neo4j graph; deployed on Azure Container Apps with Bicep IaC, Docker/GitHub Actions CI-CD, LangSmith/OTel tracing, openevals harness.

[Autonomous Data Analyst](https://github.com/trayan4/autonomous-data-analyst), <https://github.com/trayan4/autonomous-data-analyst>

Engineered a LangGraph multi-agent text-to-SQL system on Azure with governance-first routing, execution-repair validation, forced RCA decomposition, model tiering, and OpenTelemetry cost tracing; 87% on a 110-question golden set.

[Multimodal Agentic Reasoning Assistant \(MARA\)](https://github.com/trayan4/MARA-AI-Agent), <https://github.com/trayan4/MARA-AI-Agent>

Designed a multi-agent system coordinating RAG, Vision, Data & Web Search agents (3–5 parallel agent calls per query); hybrid FAISS+BM25 retrieval (recall@10 ~0.75), adaptive planning, FastAPI (p95 ~30s).

[GPT-Style Language Model from Scratch](https://github.com/trayan4/llm-from-scratch), <https://github.com/trayan4/llm-from-scratch>

Implemented a decoder-only 124M-parameter LLM from scratch in PyTorch — BPE tokenization, multi-head causal self-attention, training loop, instruction fine-tuning; trained on 124M tokens.

AWARDS

Spotlight Award, Client — delivery of the privacy-safe analytics agent platform

02/2026

Growth Catalyst Award, Accenture — technical proficiency and project delivery

12/2025